Optical Networking

SONET/SDH

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Note

SONET was developed by ANSI;
SDH was developed by ITU-T.

Synchronous Optical Network (SONET) และ
Synchronous Digital Hierarchy (SDH)
เป็นเทคโนโลยีที่ใช้ในการส่งข้อมูลปริมาณมากผ่านใยแก้วนำแสง

Concepts

SONET/SDH is a synchronous network using synchronous TDM multiplexing.

All clocks in the system are locked to a master clock.

This is used as a transport network to carry loads from other WANs.

ARCHITECTURE

The architecture of a SONET system : signals, devices, and connections.

Signals

SONET defines a hierarchy of electrical signaling levels called synchronous transport signals (STSs). Each STS level (STS-1 to STS-768) supports a certain data rate. The corresponding optical signals are called optical carriers (OCs).

SDH specifies a similar system called a synchronous transport module (STM).

Table 17.1 *SONET/SDH rates*

STS	*OC*	*Rate (Mbps)*	*STM*
STS-1	OC-1	51.840	
STS-3	OC-3	155.520	**STM-1**
STS-9	OC-9	466.560	**STM-3**
STS-12	OC-12	622.080	**STM-4**
STS-18	OC-18	933.120	**STM-6**
STS-24	OC-24	1244.160	**STM-8**
STS-36	OC-36	1866.230	**STM-12**
STS-48	OC-48	2488.320	**STM-16**
STS-96	OC-96	4976.640	**STM-32**
STS-192	OC-192	9953.280	**STM-64**
STS-768	***OC-768***	***39813.12 Mbps***	***STM-256***
STS-3072	*OC-3072*	*159252.48 Mbps*	*STM-1024*

ARCHITECTURE

Devices

Figure 17.1 shows a simple link using SONET devices. SONET transmission relies on three basic devices: STS multiplexers/demultiplexers, regenerators, add/drop multiplexers and terminals.

Figure 17.1 *A simple network using SONET equipment*

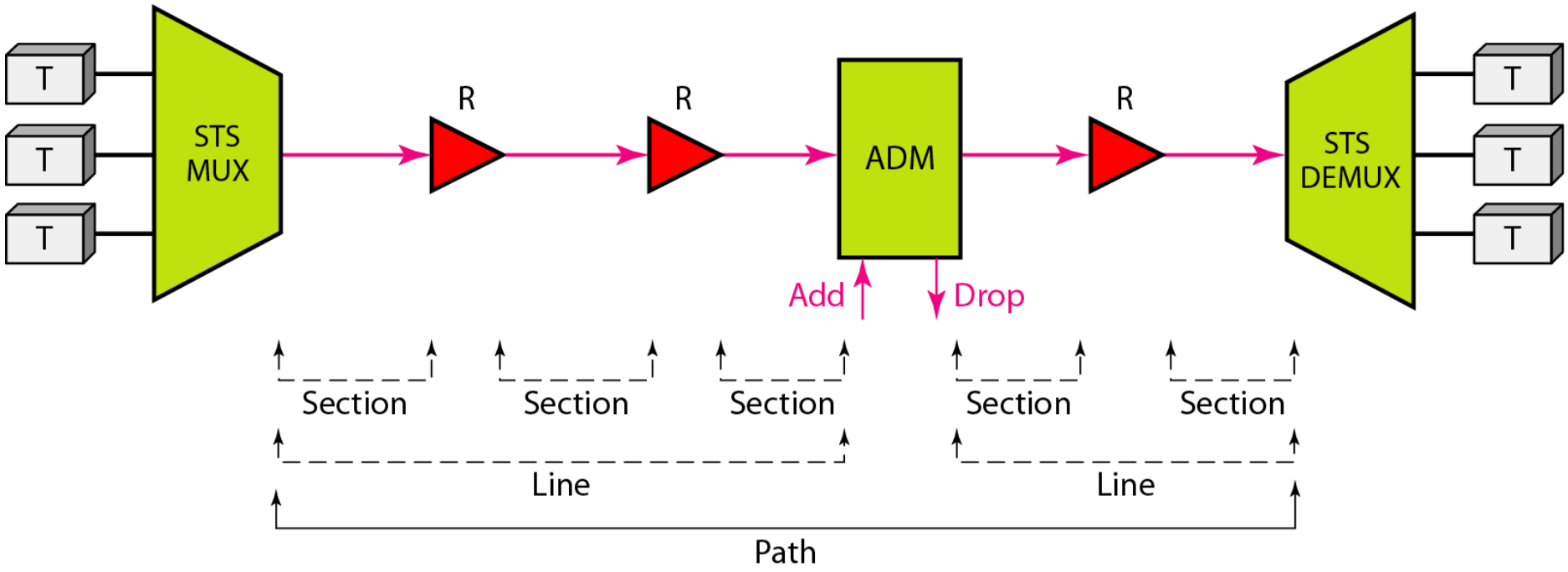
ADM: Add/drop multiplexer

STS MUX: Synchronous transport signal multiplexer

STS DEMUX: Synchronous transport signal demultiplexer

R: Regenerator

T: Terminal



ARCHITECTURE

STS Multiplexer/Demultiplexer

They provide the interface between an electrical tributary network and the optical network. An STS multiplexer multiplexes signals from multiple electrical sources and creates the corresponding OC signal. An STS demultiplexer demultiplexes an optical OC signal into corresponding electric signals.

Regenerator

A regenerator is a repeater that takes a received optical signal *(OC-n),* demodulates it into the corresponding electric signal *(STS-n),* regenerates the electric signal, and finally modulates the electric signal into its correspondent *OC-n* signal.

ARCHITECTURE

Add/drop Multiplexer

Add/drop multiplexers allow insertion and extraction of signals. An **add/drop multiplexer(ADM)** can add STSs coming from different sources into a given path or can remove a desired signal from a path and redirect it without demultiplexing the entire signal.

Terminals

For example, in the Internet, a terminal can be a router that needs to send packets to another router at the other side of a SONET network.

ARCHITECTURE

Connections

The devices defined in the previous section are connected using *sections, lines,* and *paths*.

Sections

A section is the optical link connecting two neighbor devices: multiplexer to multiplexer, multiplexer to regenerator, or regenerator to regenerator.

Lines

A line is the portion of the network between two multiplexers: STS multiplexer to add/drop multiplexer.

Paths

A path is the end-to-end portion of the network between two STS multiplexers.

SONET LAYERS

Note

SONET defines four layers: path, line, section, and photonic.

They correspond to both the physical and the data link layers.

SONET LAYERS

Path Layer

The path layer is responsible for a signal from optical source to optical destination.

Line Layer

The line layer is responsible for the movement of a signal across a physical line.

Section Layer

The section layer is responsible for the movement of a signal across a physical section.

Photonic Layer

The photonic layer corresponds to the physical layer of the OSI model.

Figure 17.2 *SONET layers compared with OSI or the Internet layers*

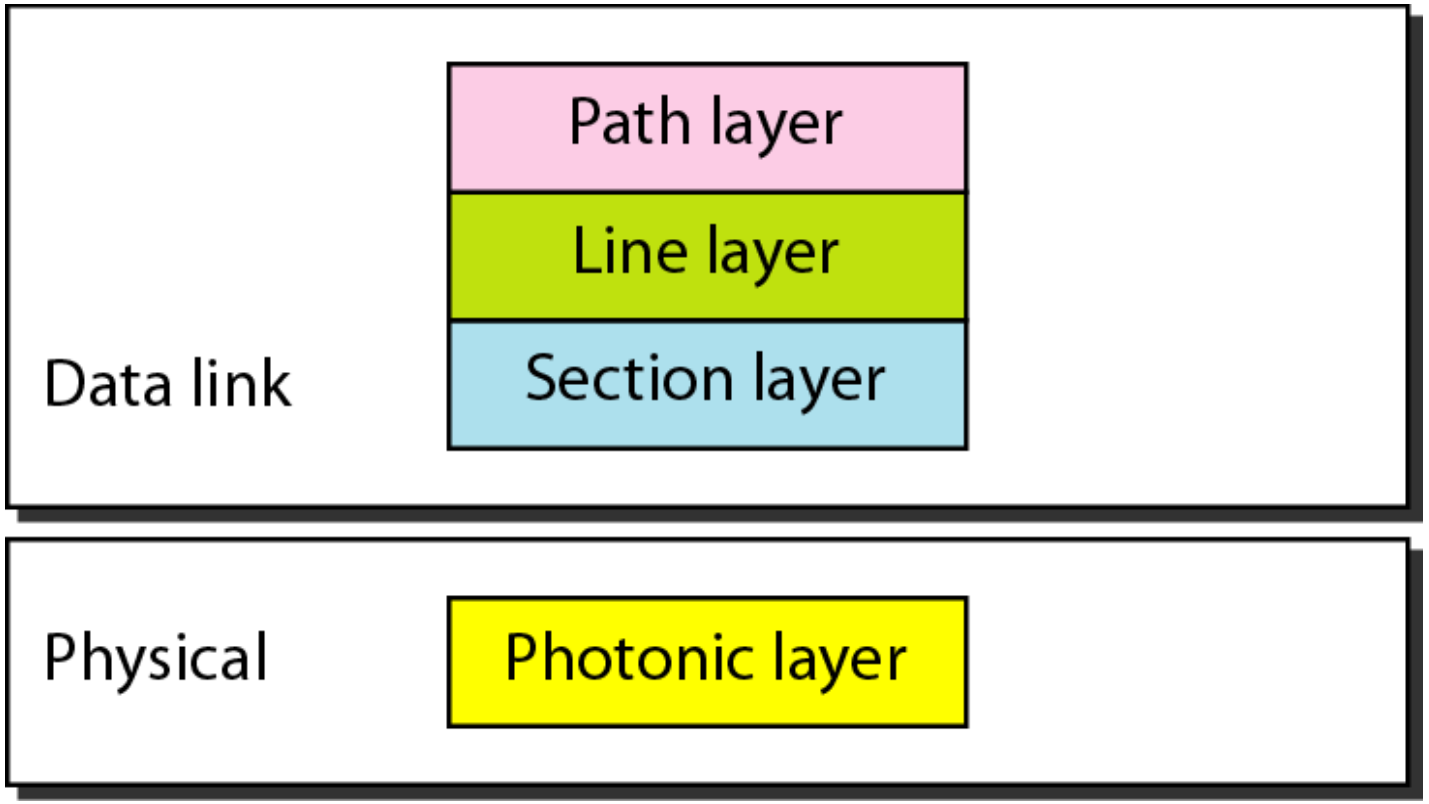
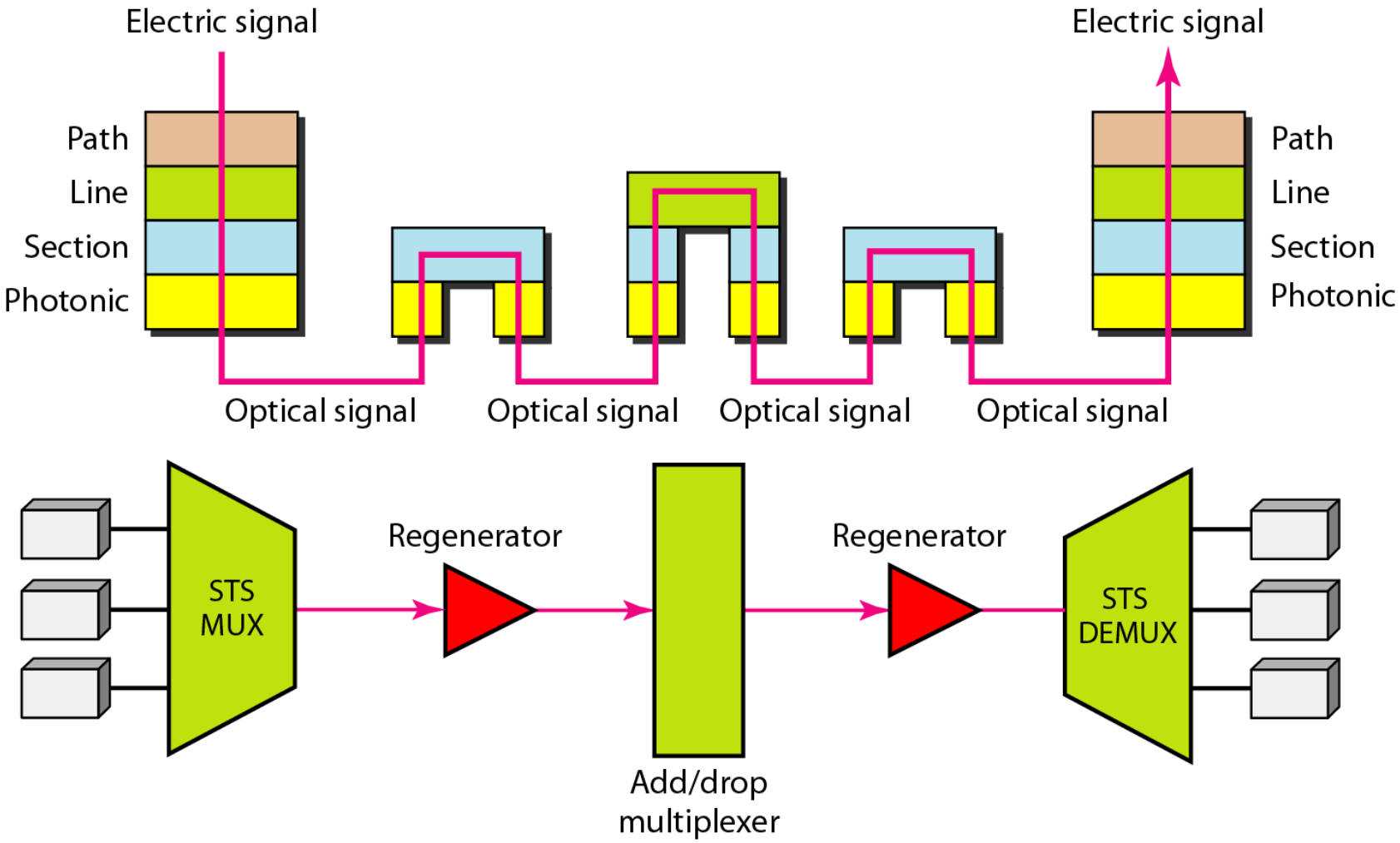


Figure 17.3 *Device–layer relationship in SONET*

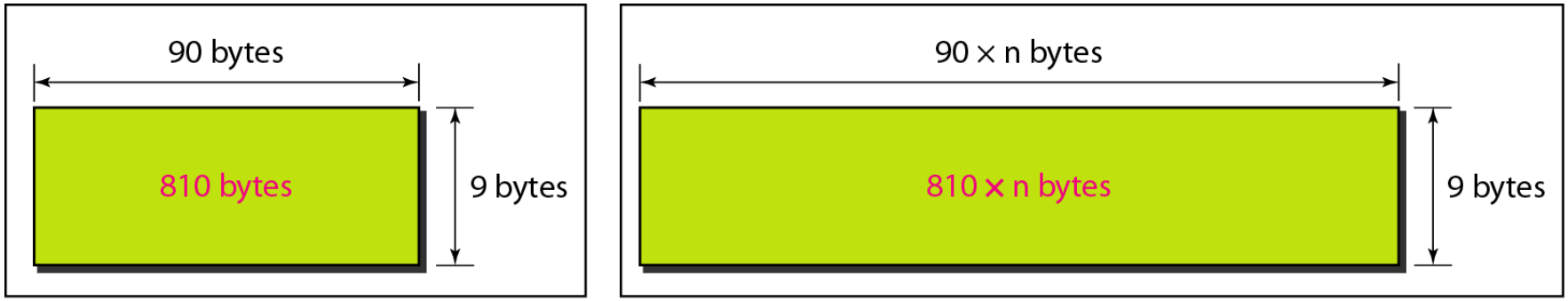


SONET FRAMES

Each synchronous transfer signal STS-n is composed of 8000 frames. Each frame is a two-dimensional matrix of bytes with 9 rows by 90 × n columns.

For example, STS-1 frame is 9 rows by 90 columns (810 bytes), and an STS-3 is 9 rows by 270 columns (2430 bytes). Figure 17.4 shows the general format of an STS-1 and an *STS-n.*

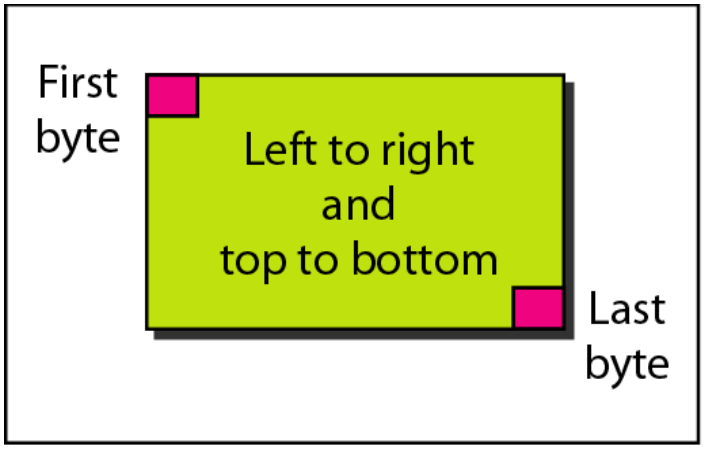
Figure 17.4 *An STS-1 and an STS-n frame*



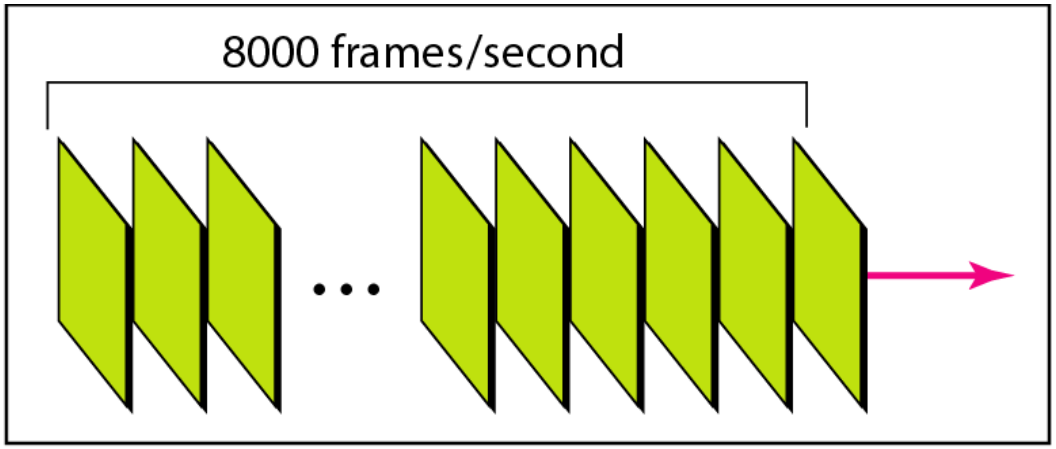
a. STS-1 frame

b. STS-n frame

Figure 17.5 *STS-1 frames in transmission*



a. Byte transmission



b. Frame transmission

SONET FRAMES

For each frame the bytes are transmitted from the left to the right, top to the bottom. For each byte, the bits are transmitted from the most significant to the least significant (left to right).

An STS-1 signal can carry 774 voice channels simultaneously (810 minus required bytes for overhead).

Note

A SONET STS-n signal is transmitted at 8000 frames per second.

Note

Each byte in a SONET frame can carry a digitized voice channel.

Example 1

Find the data rate of an STS-1 signal.

Solution

STS-1, like other STS signals, sends 8000 frames per second. Each STS-1 frame is made of 9 by (1×90) bytes. Each byte is made of 8 bits. The data rate is

Example 2

Find the data rate of an STS-3 signal.

Solution

STS-3, like other STS signals, sends 8000 frames per second. Each STS-3 frame is made of 9 by (3×90) bytes. Each byte is made of 8 bits. The data rate is

Note

In SONET, the data rate of an STS-n signal is *n* times the data rate of an STS-1 signal.

Note

In SONET,
the duration of any frame is 125 µs.

STS-1 Frame Format

The basic format of an STS-1 frame is shown in Figure 17.6. As before, a SONET frame is a matrix of 9 rows of 90 bytes (octets) each, for a total of 810 bytes.

The first three columns of the frame are used for section and line overhead. The upper three rows of the first three columns are used for section overhead (SOH). The lower six are line overhead (LOH). The rest of the frame is called the synchronous payload envelope (SPE). It contains user data and path overhead (POH) needed at the user data level.

Figure 17.6 *STS-1 frame overheads*

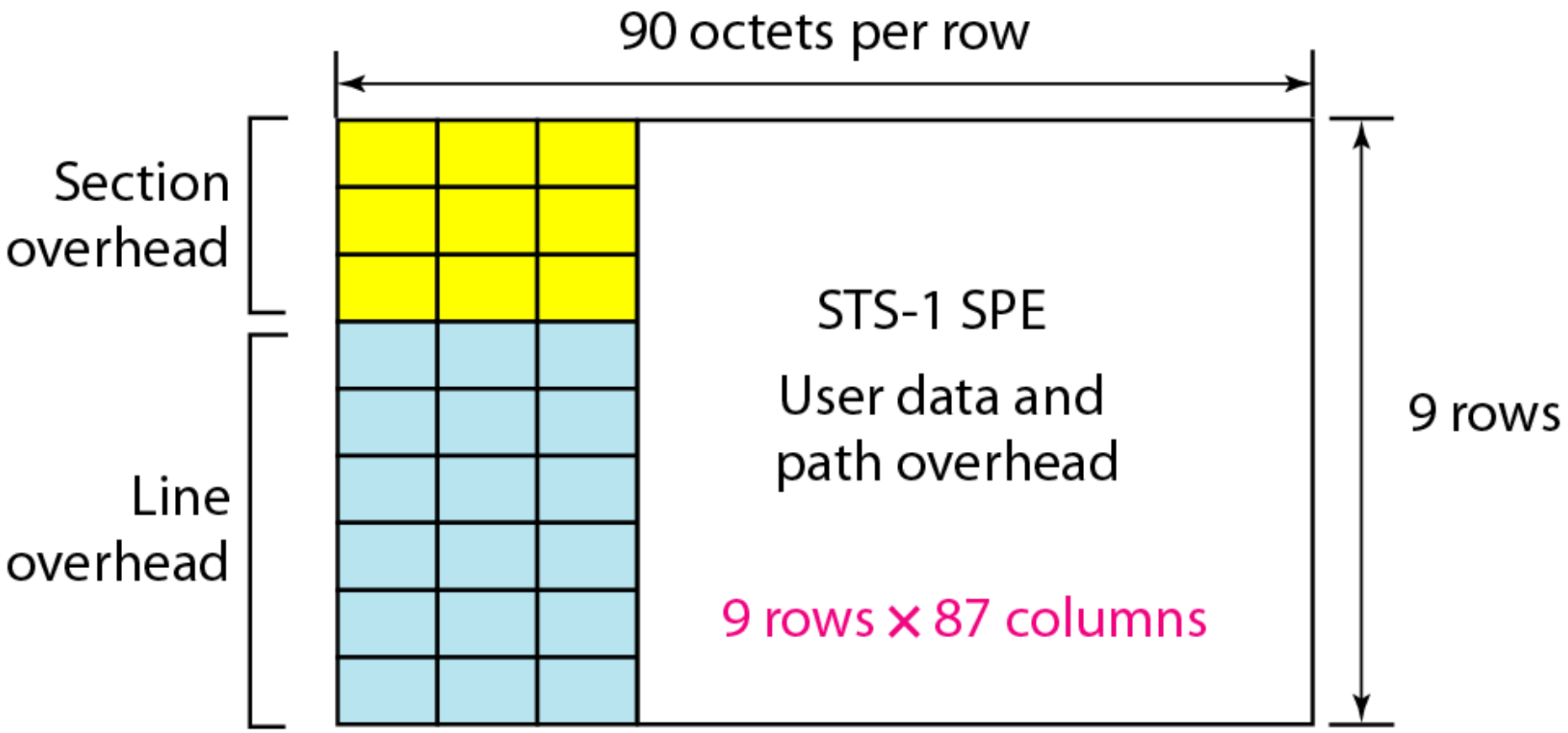
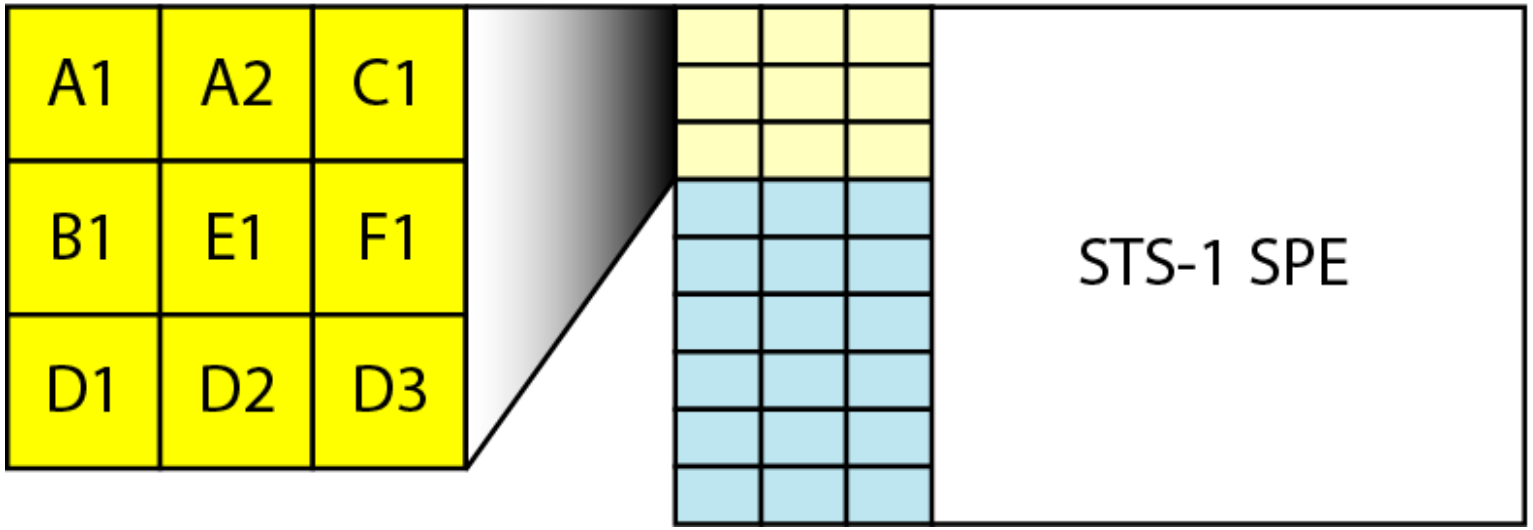


Figure 17.7 *STS-1 frame: section overhead*

A1, A2: Alignment
B1: Parity byte
C1: Identification
D1, D2, D3: Management
E1: Order wire byte
F1: User



Note

Section overhead is recalculated for each SONET device (regenerators and multiplexers).

Figure 17.8 *STS-1 frame: line overhead*

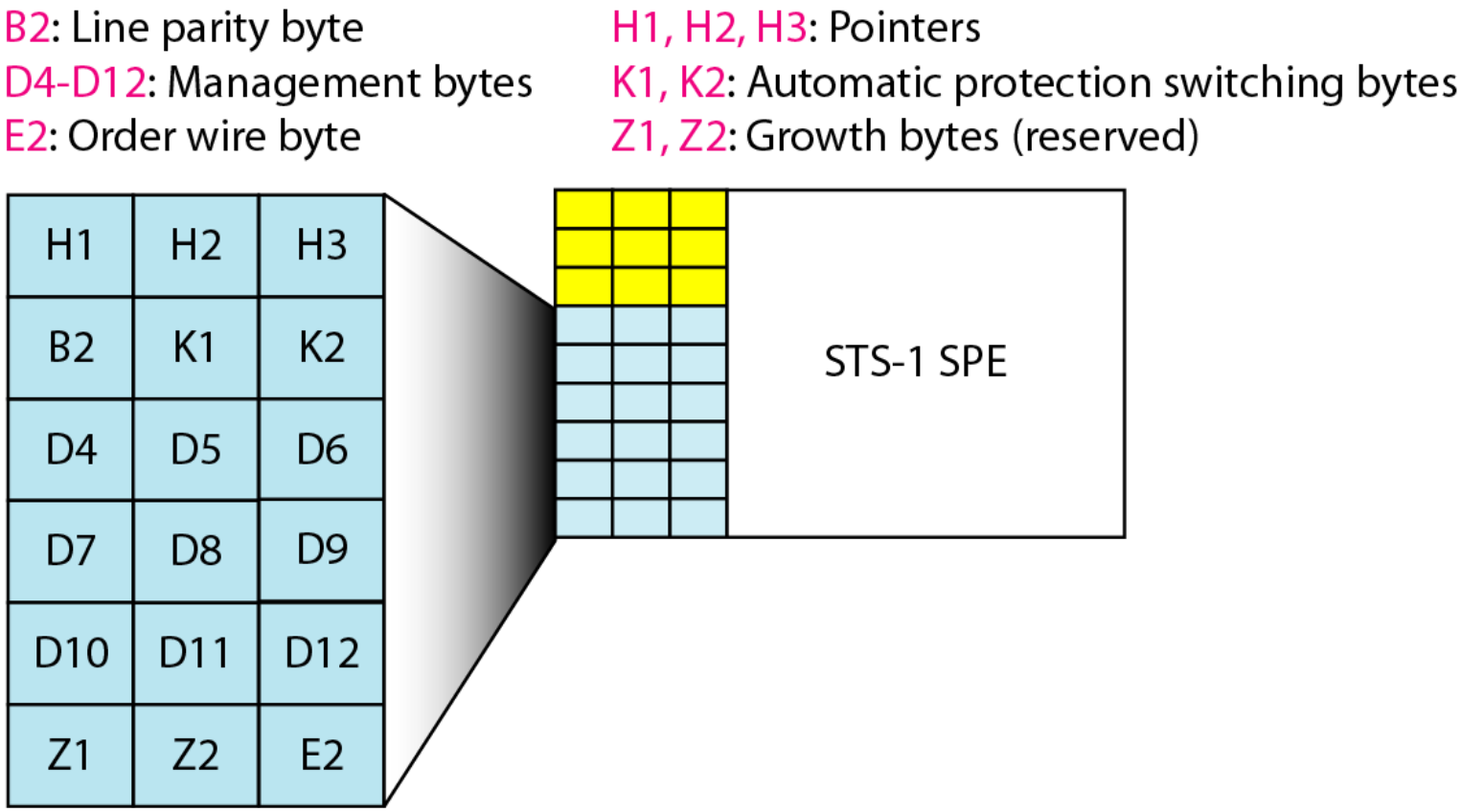
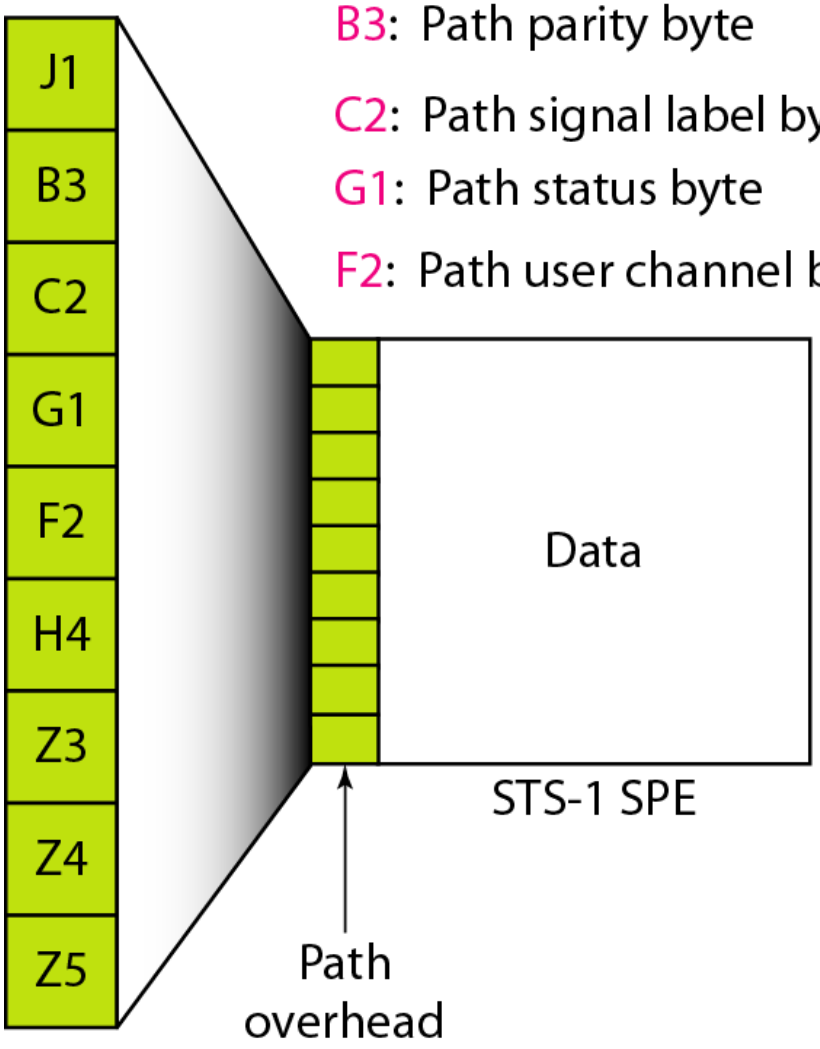


Figure 17.9 *STS-1 frame: path overhead*



B3: Path parity byte

C2: Path signal label byte

G1: Path status byte

F2: Path user channel byte

H4: Virtual tributary indicator

J1: Path trace byte

Z3, Z4, Z5: Growth bytes (reserved)

Note

Path overhead is only calculated for end-to-end (at STS multiplexers).

Example 3

What is the user data rate of an STS-1 frame and STM-1 frame (without considering the overheads)?

Solution

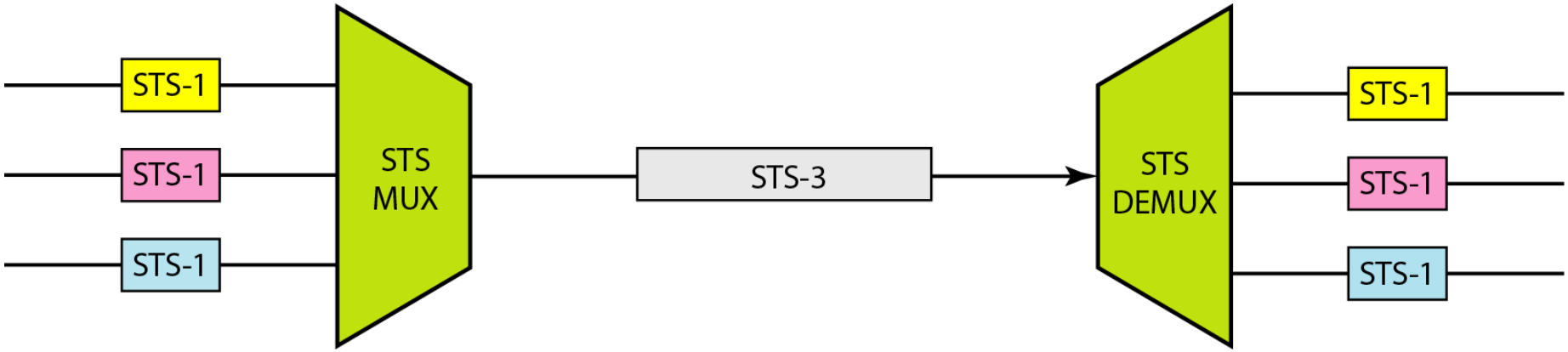
The user data part in an STS-1 frame is made of 9 rows and 86 columns. So we have

STS MULTIPLEXING

In SONET, frames of lower rate can be synchronously time-division multiplexed into a higher-rate frame. For example, three STS-1 signals (channels) can be combined into one STS-3 signal (channel), four STS-3s can be multiplexed into one STS-12, and so on.

Byte Interleaving
Concatenated Signal
Add/Drop Multiplexer

Figure 17.12 *STS multiplexing/demultiplexing*



Note

In SONET, all clocks in the network are locked to a master clock.

Byte Interleaving

Synchronous TDM multiplexing in SONET is achieved by using byte interleaving.

For example, when three STS-1 signals are multliplexed into one STS-3 signal, each set of 3 bytes in the STS-3 signal is associated with 1 byte from each STS-1 signal.

Byte interleaving also preserves the corresponding section and line overhead as shown in Figure 17.14. As the figure shows, the section overheads from three STS-1 frames are interleaved together to create a section overhead for an STS-1 frame.

Figure 17.13 *Byte interleaving*

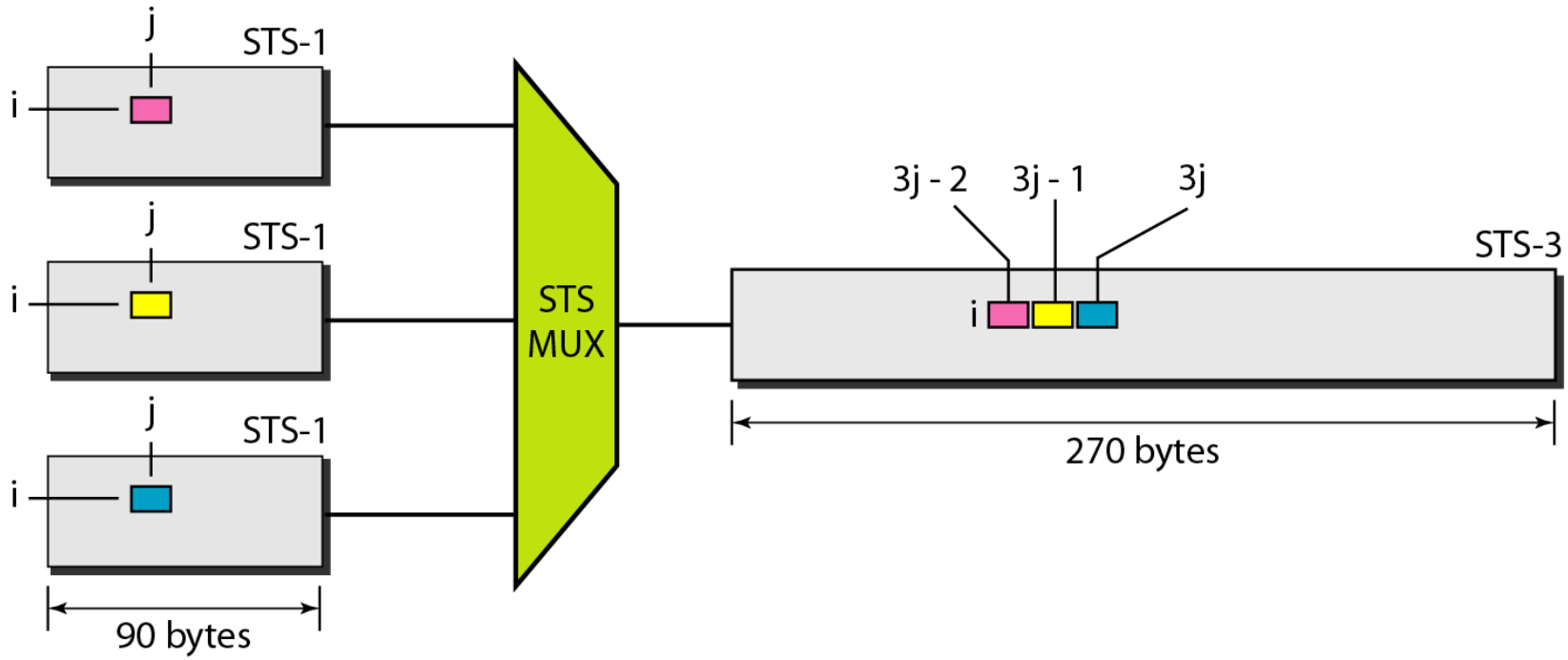
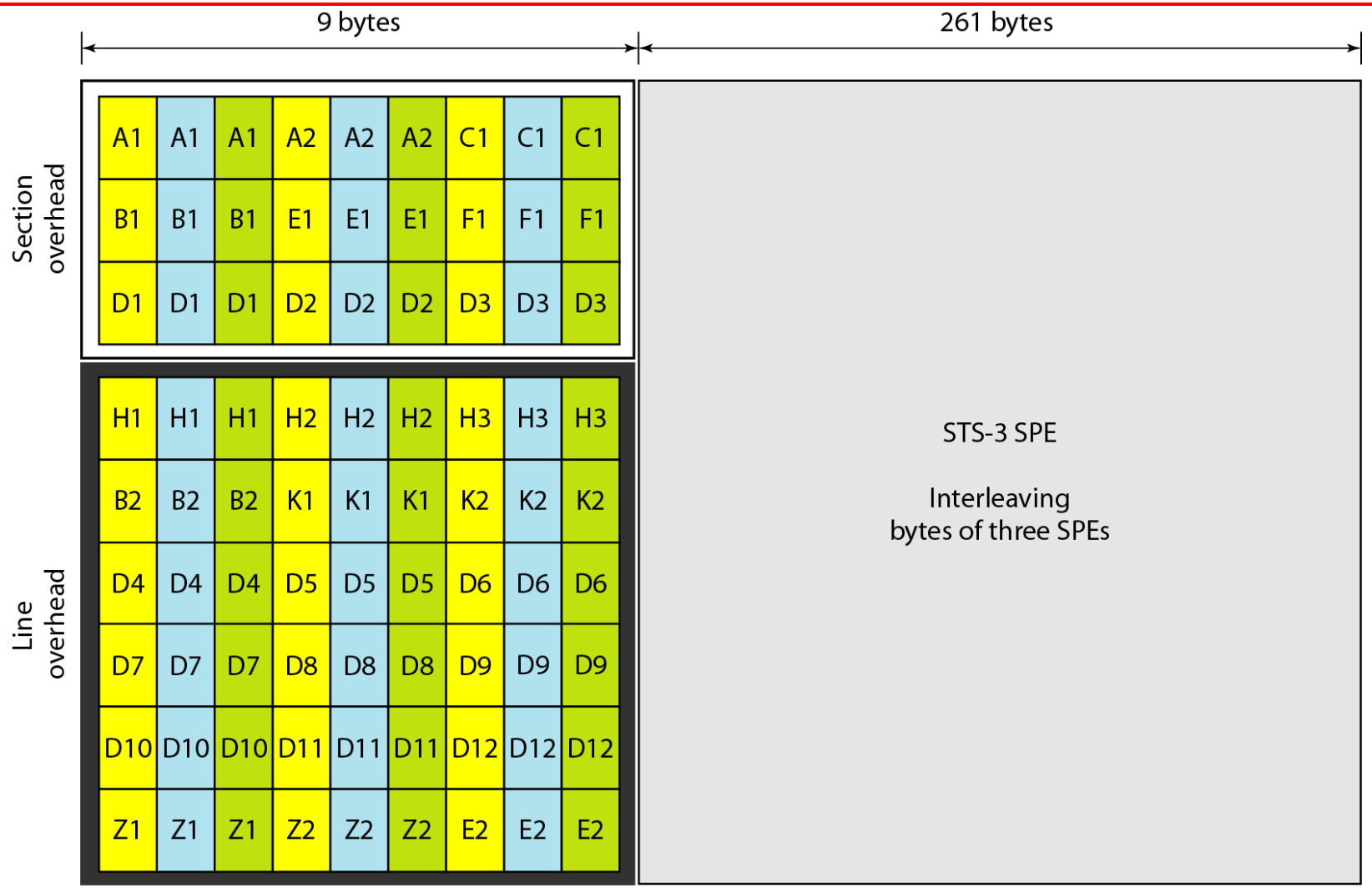


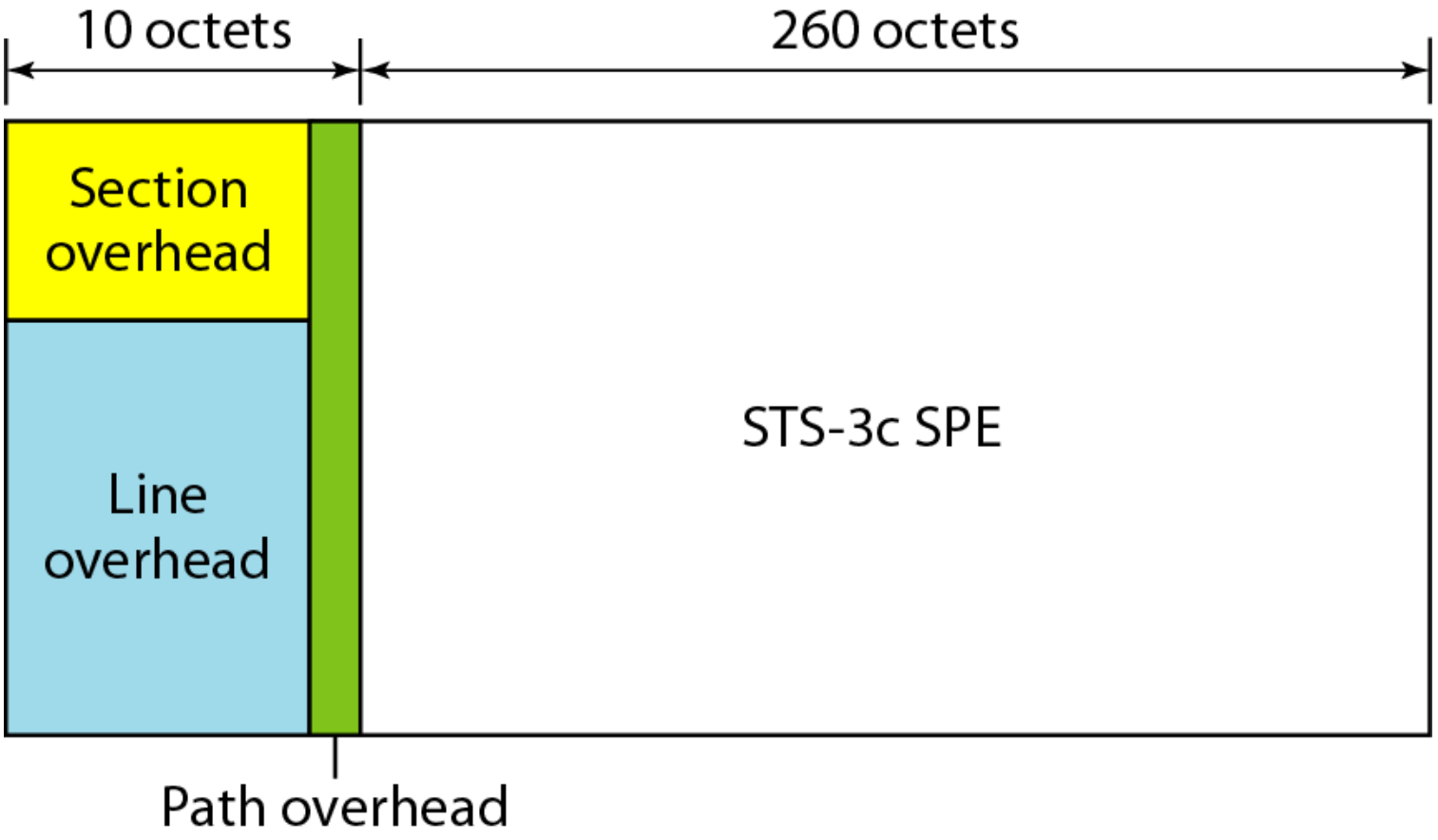
Figure 17.14 *An STS-3 frame*



Concatenated Signal

Sometimes, we have a signal with one *STS-n* signal (channel) that cannot be demultiplexed into *n* STS-1 signals. To specify that the signal cannot be demultiplexed, the suffix c (for concatenated) is added to the name of the signal.

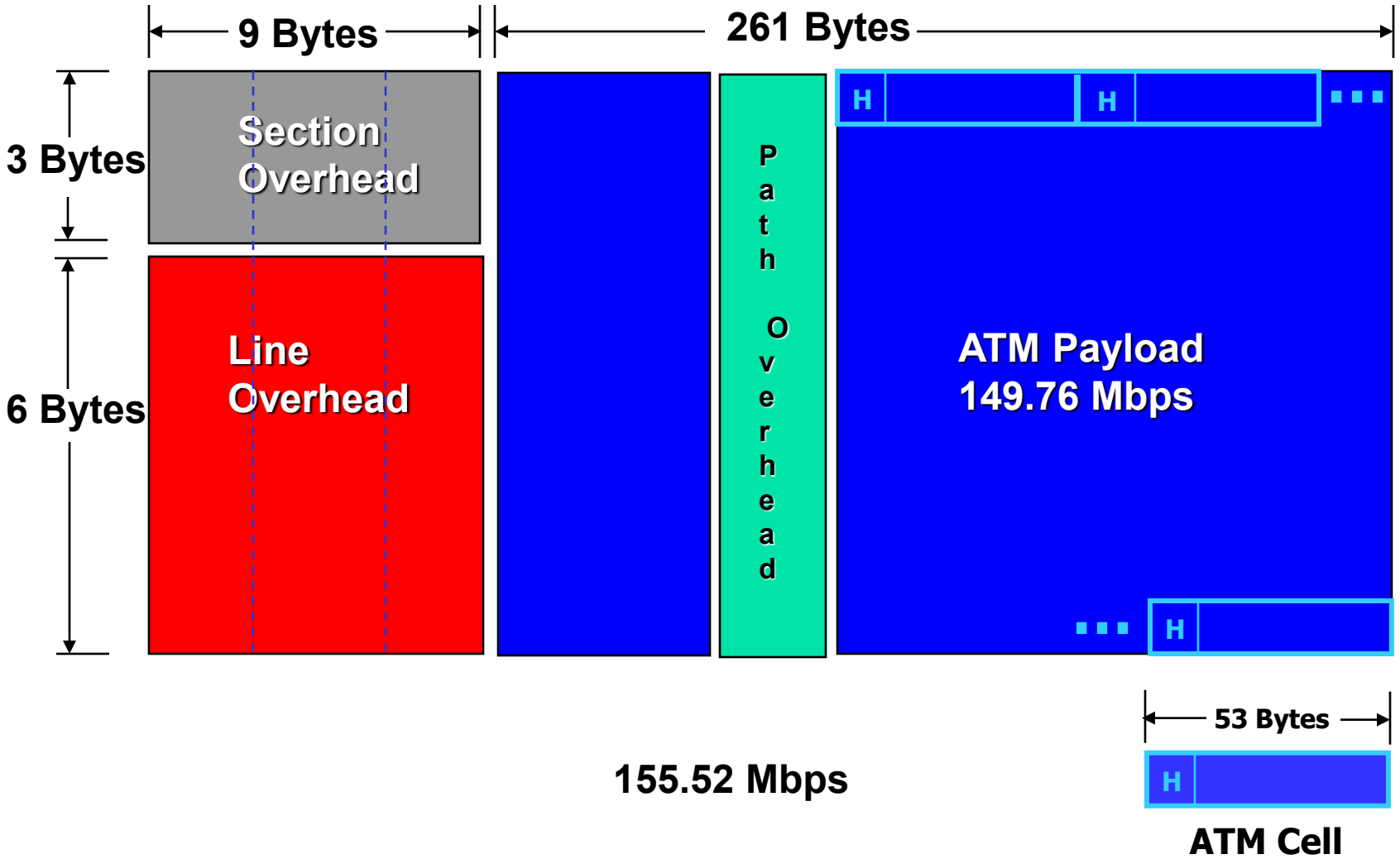
Figure 17.15 *A concatenated STS-3c signal*



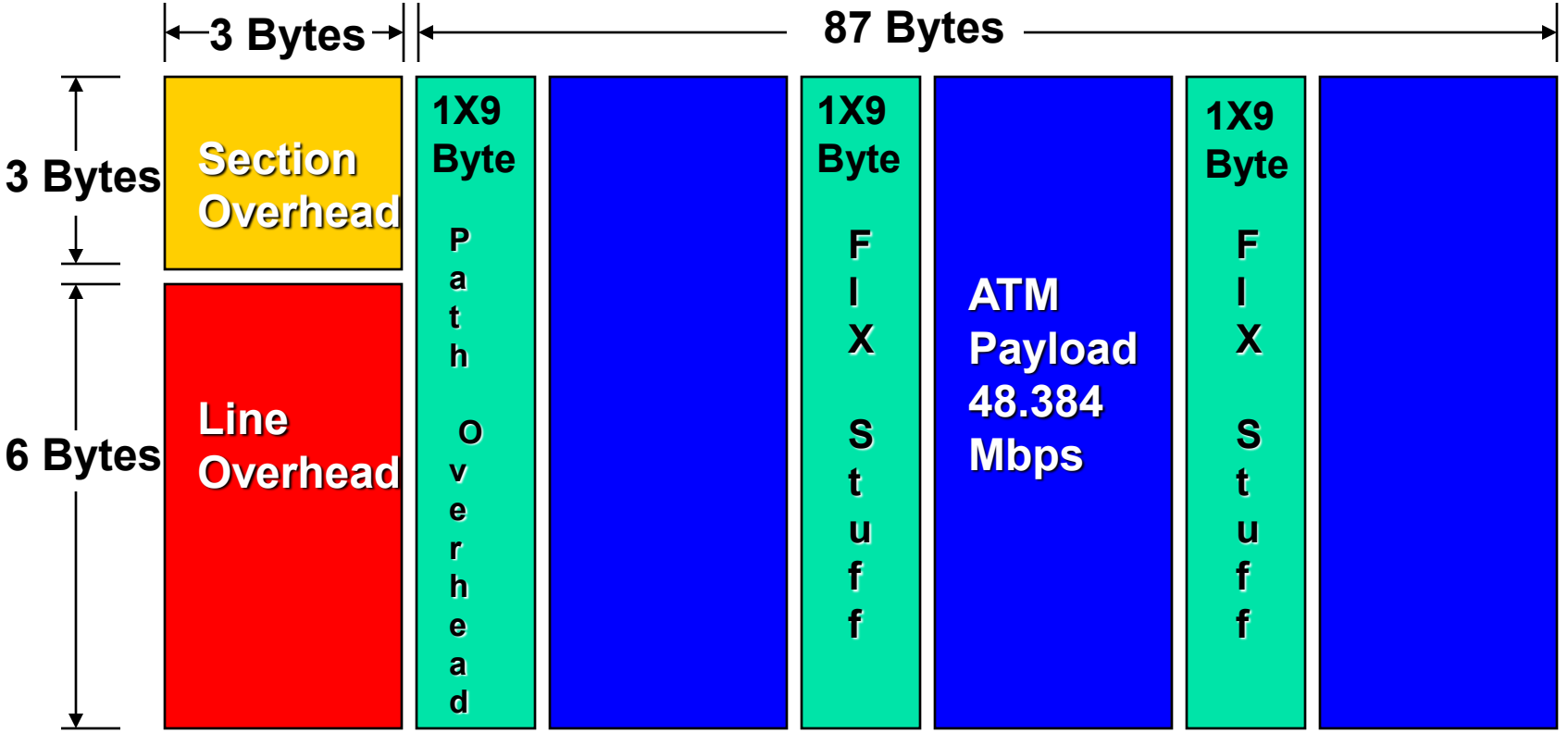
Note

An STS-3c signal can carry 44 ATM cells as its SPE.

STS-3c Frame Structure



STS- 1 Frame with ATM

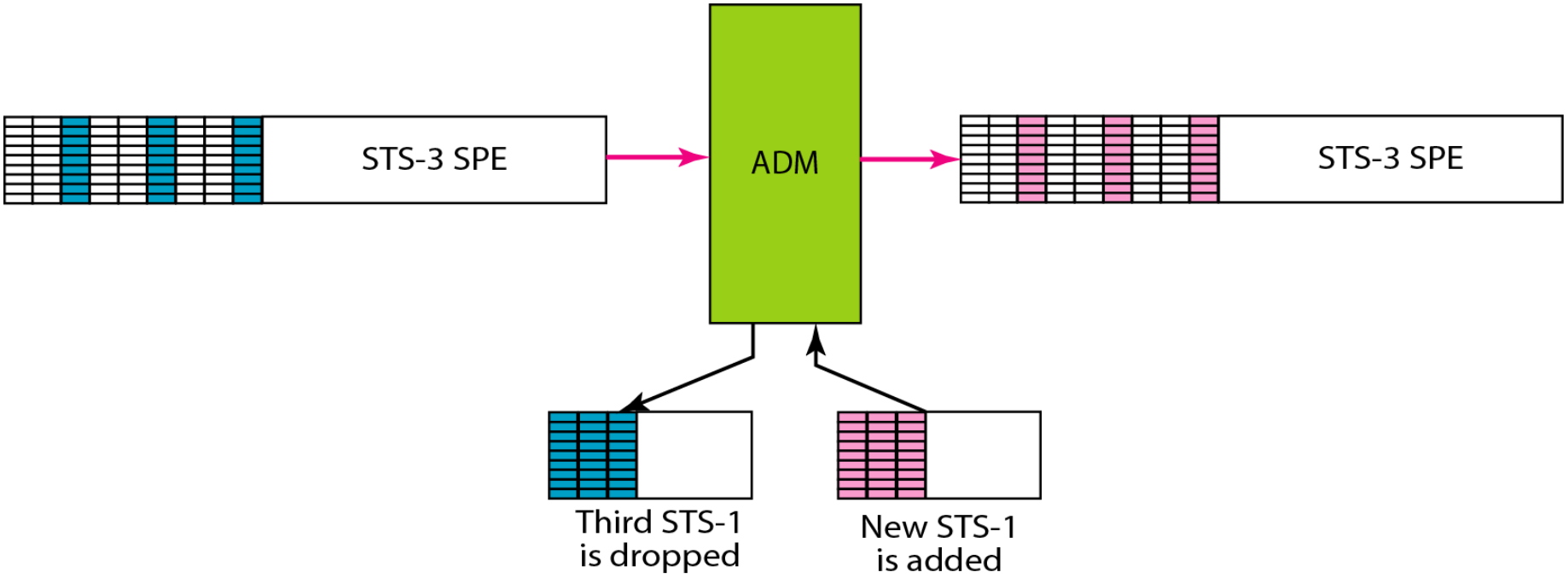


51.84 Mbps

Add/Drop Multiplexer

An add/drop multiplexer operates at the line layer. The add/drop multiplexer (ADM) only removes the corresponding bytes and replaces them with the new bytes (including the bytes in the section and line overhead).

Figure 17.16 *Dropping and adding STS-1 frames in an add/drop multiplexer*



VIRTUAL TRIBUTARIES

SONET is designed to carry broadband payloads. Current digital hierarchy data rates, however, are lower than STS-1. To make SONET backward-compatible with the current hierarchy, its frame design includes a system of virtual tributaries (VTs). A virtual tributary is a partial payload that can be inserted into an STS-1.

Types of VTs

Figure 17.25 *Virtual tributaries*

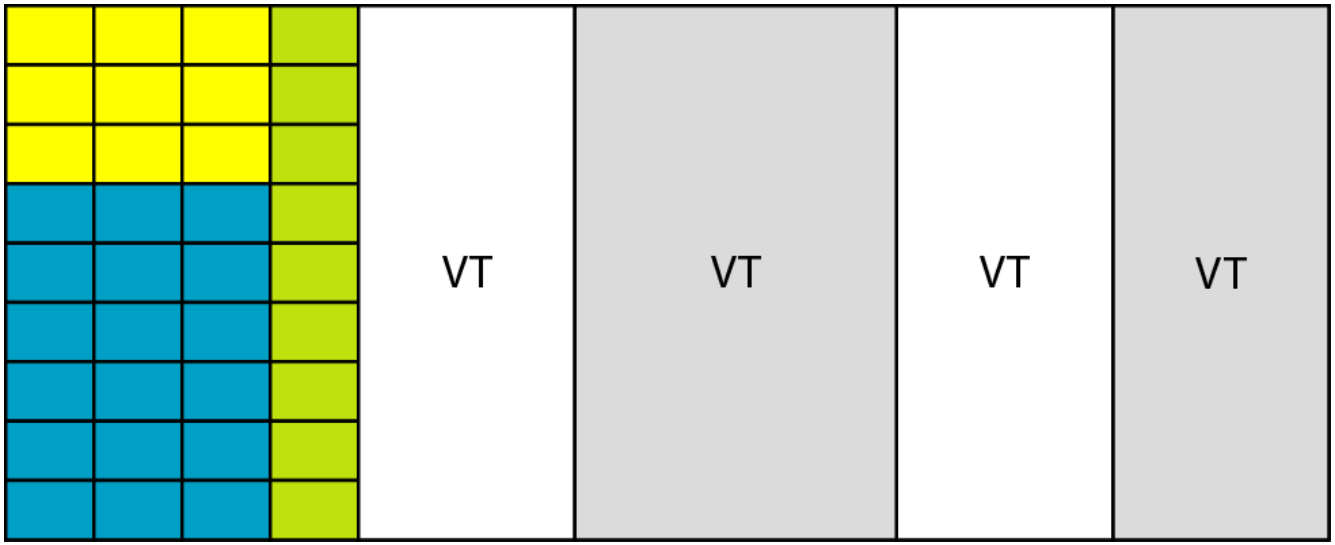
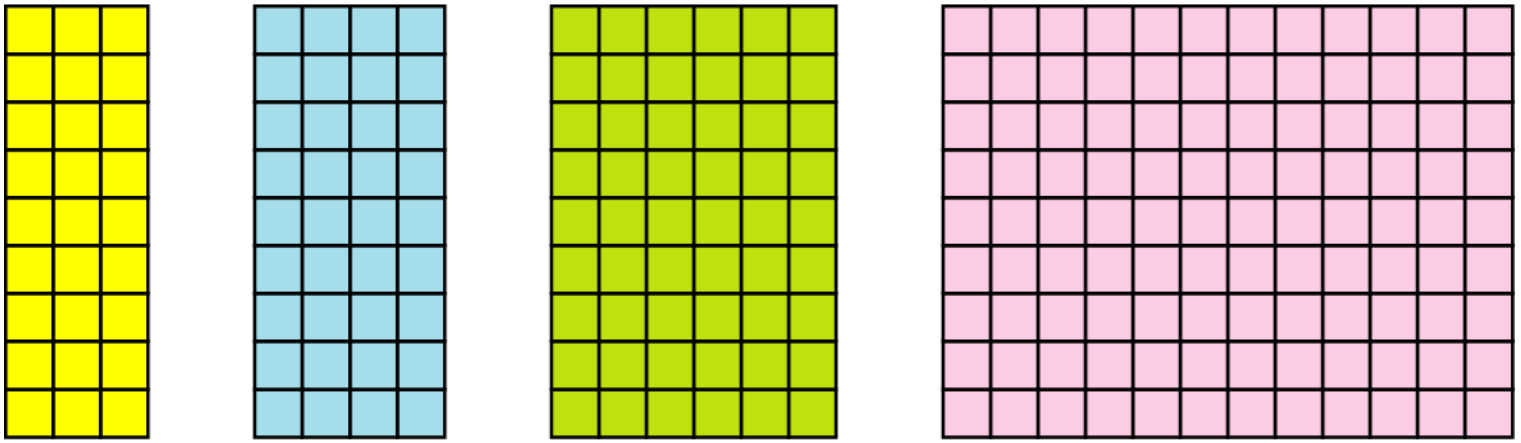
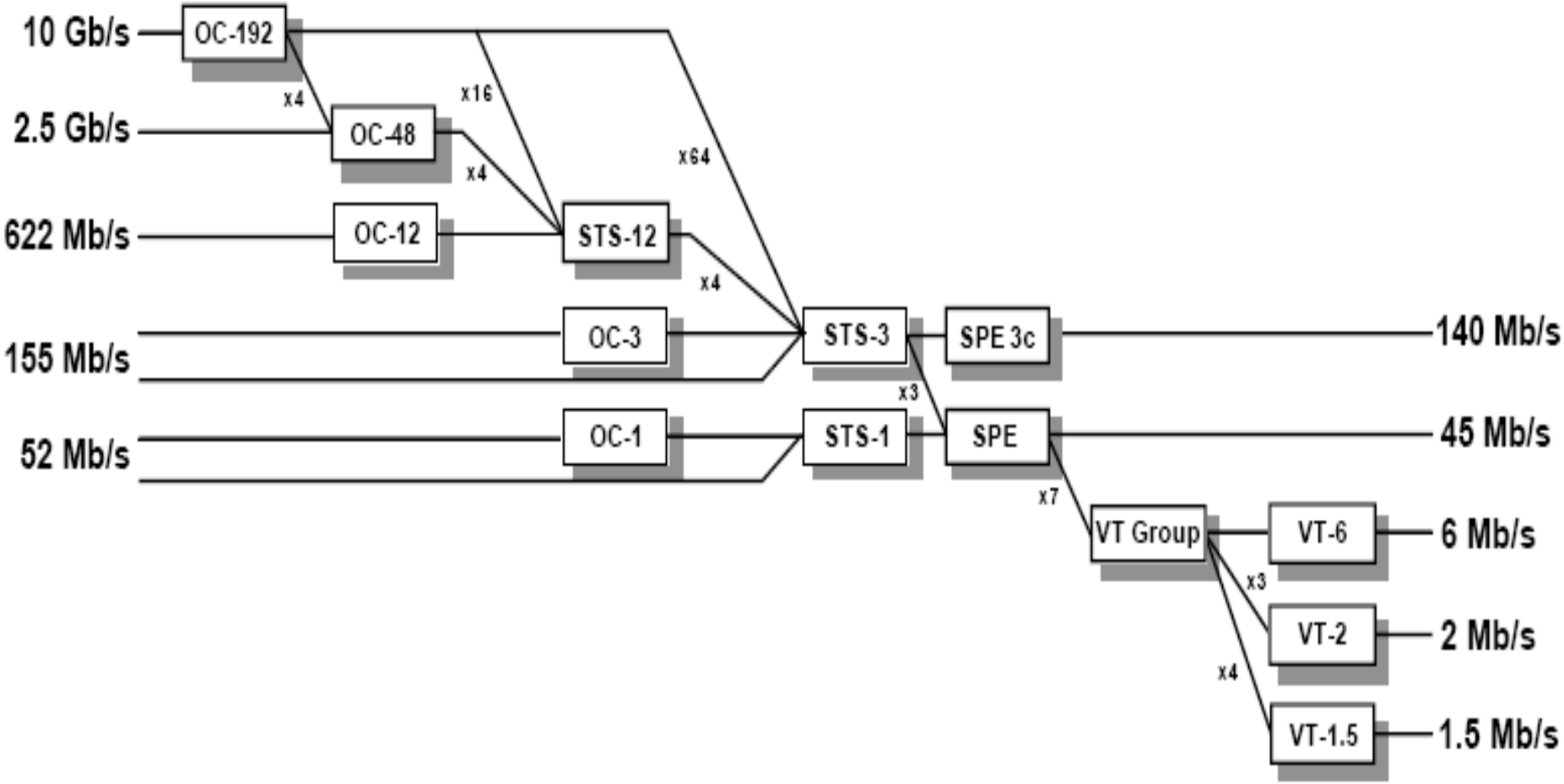


Figure 17.26 *Virtual tributary types*

VT1.5 = 8000 frames/s 3 columns 9 rows 8 bits = 1.728 Mbps
VT2 = 8000 frames/s 4 columns 9 rows 8 bits = 2.304 Mbps
VT3 = 8000 frames/s 6 columns 9 rows 8 bits = 3.456 Mbps
VT6 = 8000 frames/s 12 columns 9 rows 8 bits = 6.912 Mbps



SONET multiplexing Hierarchy



- All services below DS3 rate are transported in the VT structure.

SONET NETWORKS

Using SONET equipment, we can create a SONET network that can be used as a high-speed backbone carrying loads from other networks *such as optical data transmission, 5G networks and so on.*

We can roughly divide SONET networks into three categories: linear, ring, and mesh networks.

Linear Networks
Ring Networks
Mesh Networks

Figure 17.17 *Taxonomy of SONET networks*

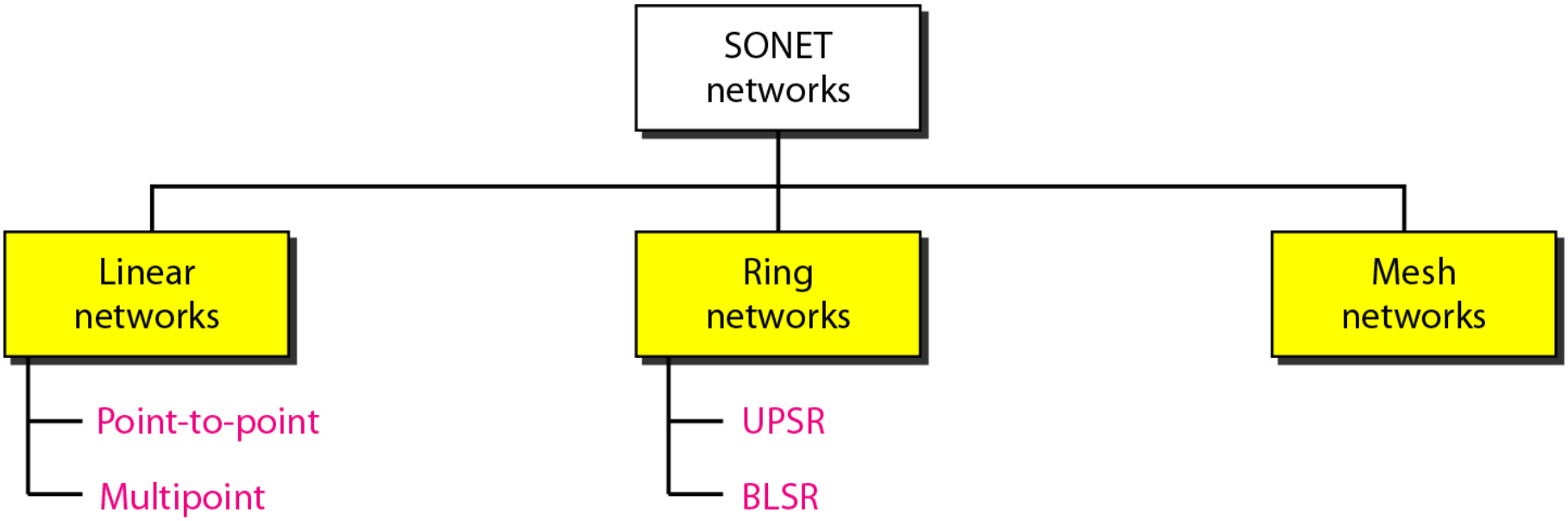


Figure 17.18 *A point-to-point SONET network*

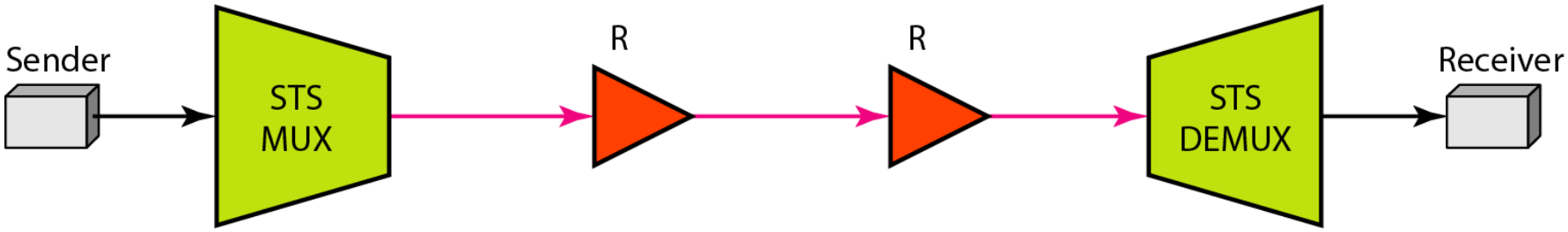


Figure 17.19 *A multipoint SONET network*

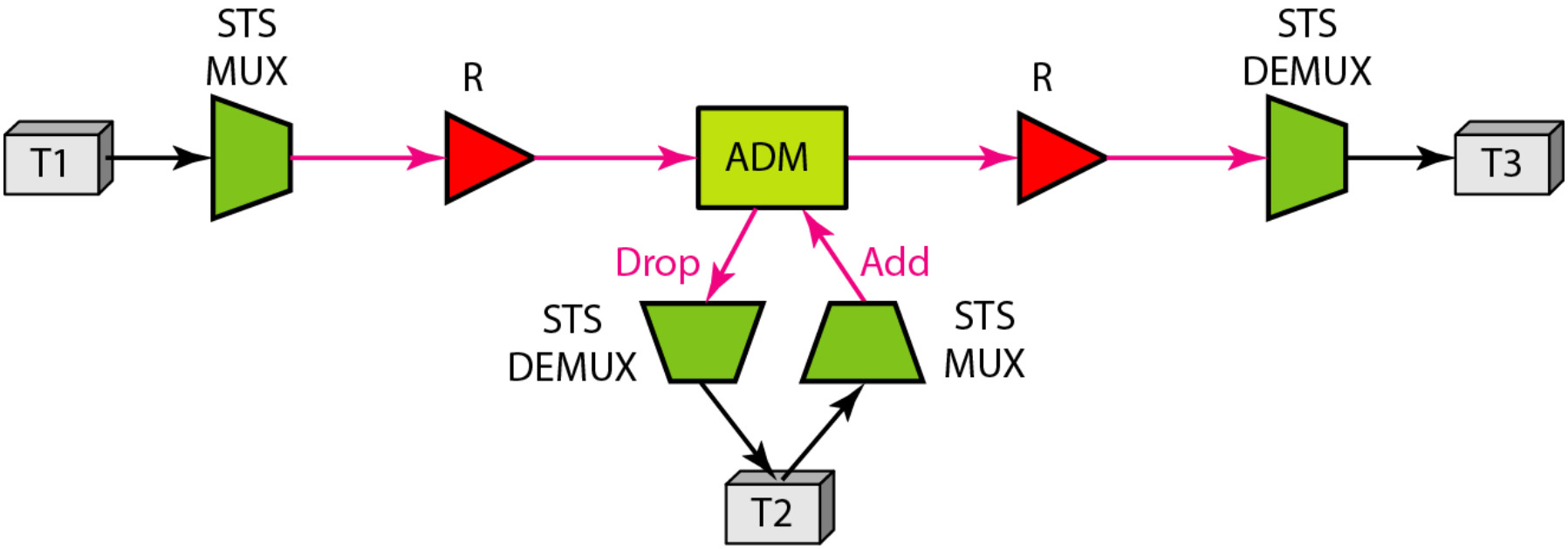
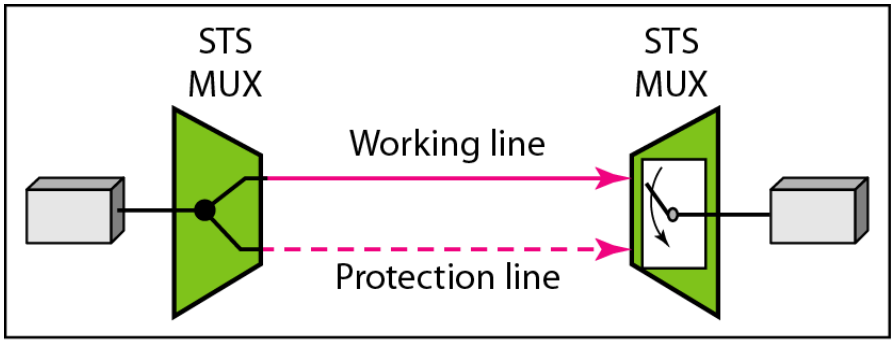
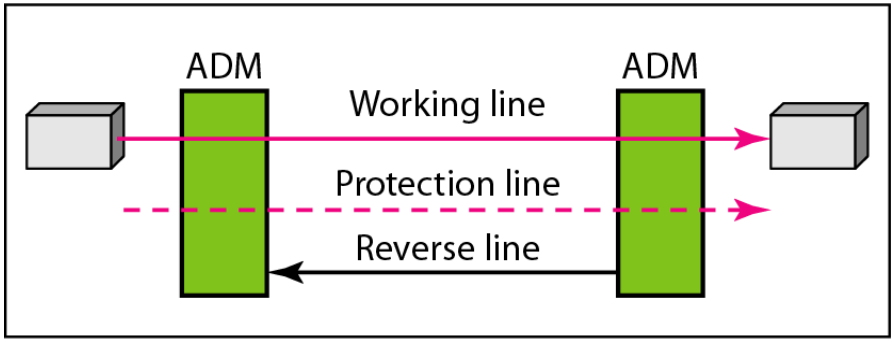


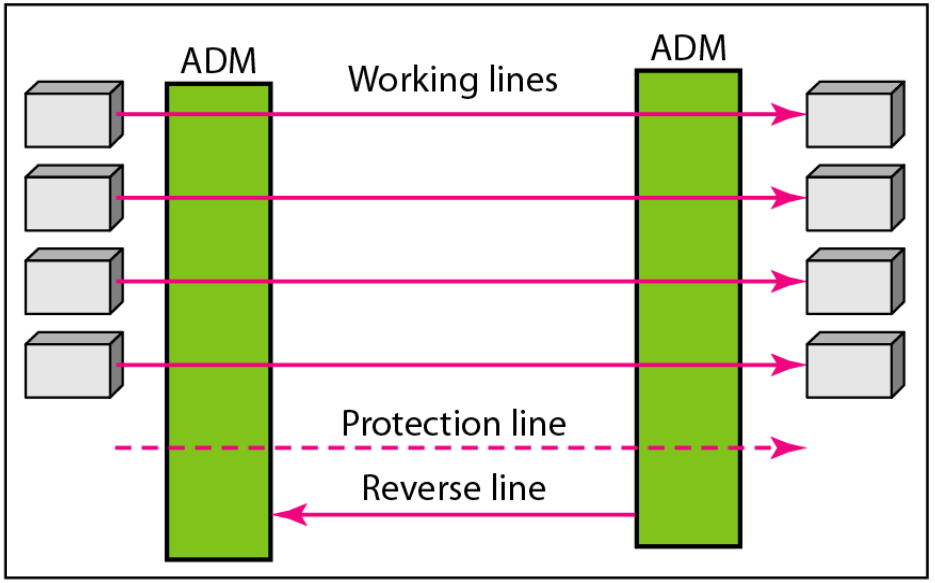
Figure 17.20 *Automatic protection switching in linear networks*



a. One-plus-one APS



b. One-to-one APS



c. One-to-many APS

Figure 17.21 *A unidirectional path switching ring*

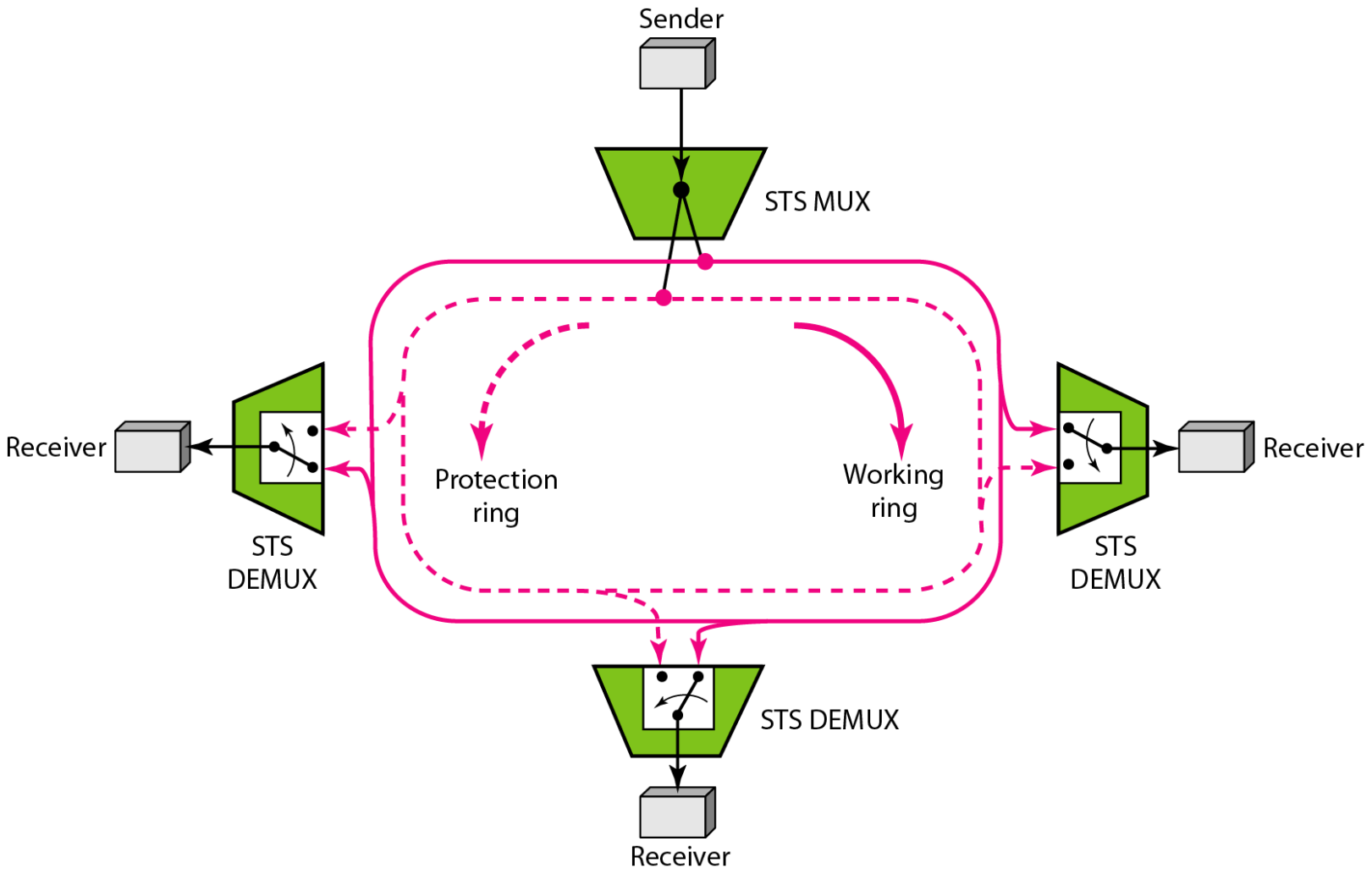


Figure 17.22 *A bidirectional line switching ring*

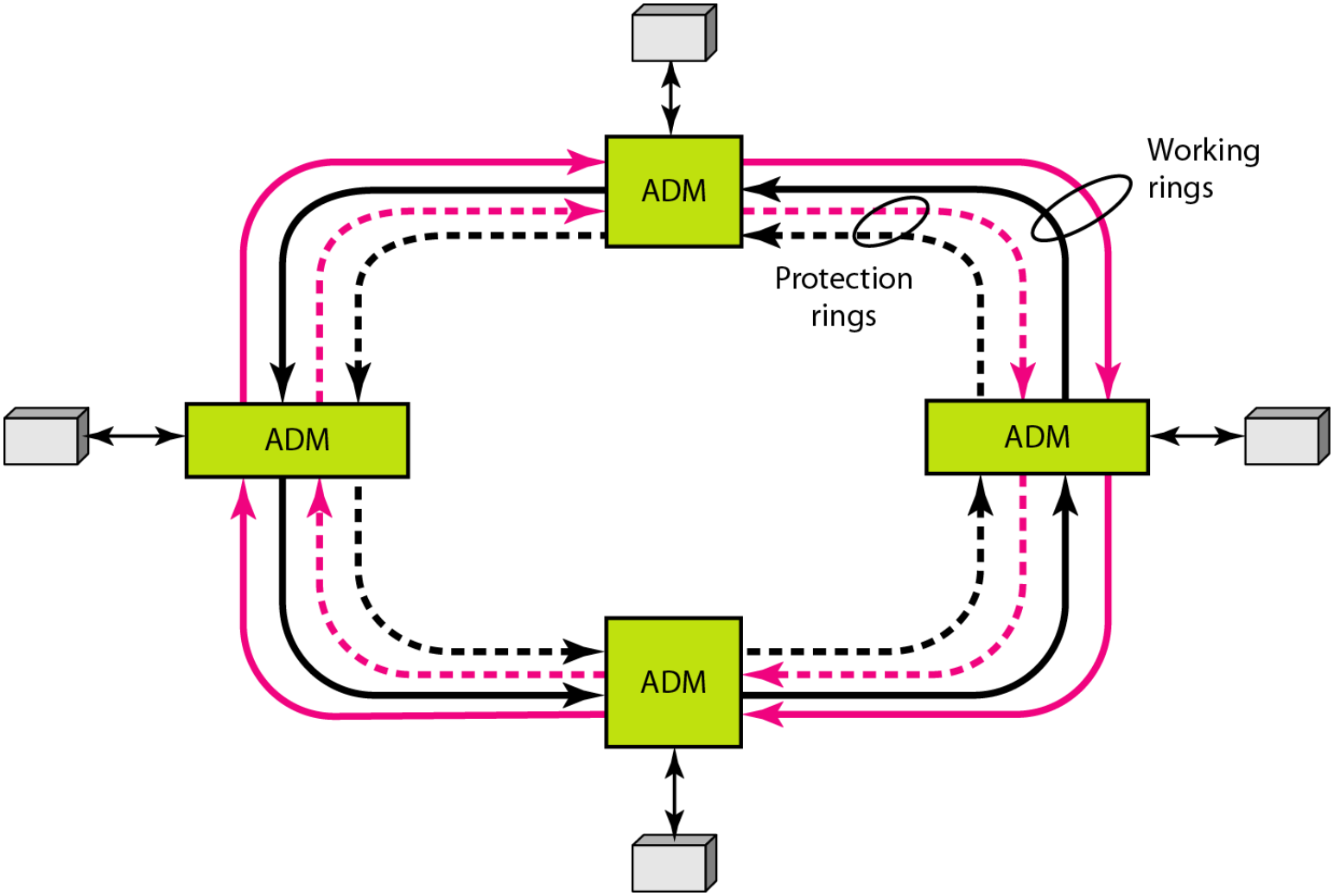


Figure 17.23 *A combination of rings in a SONET network*

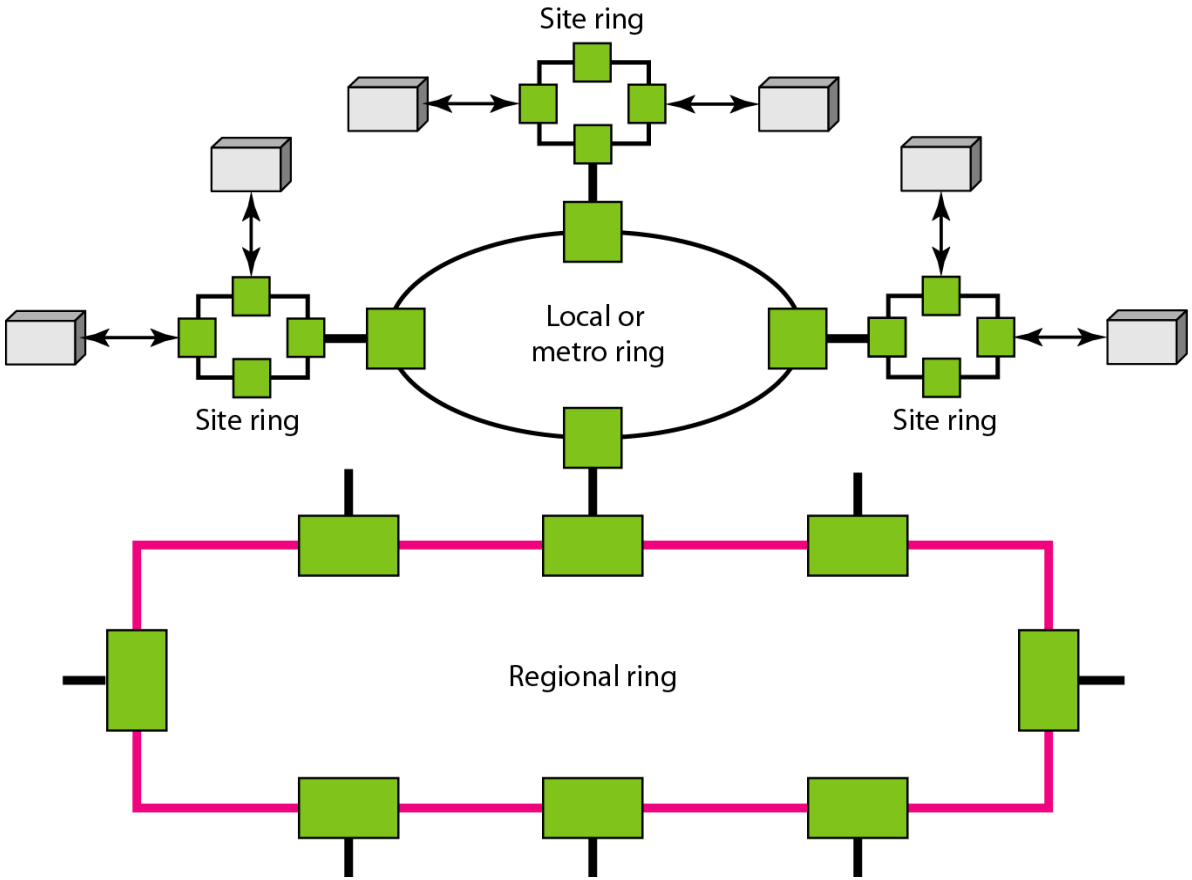
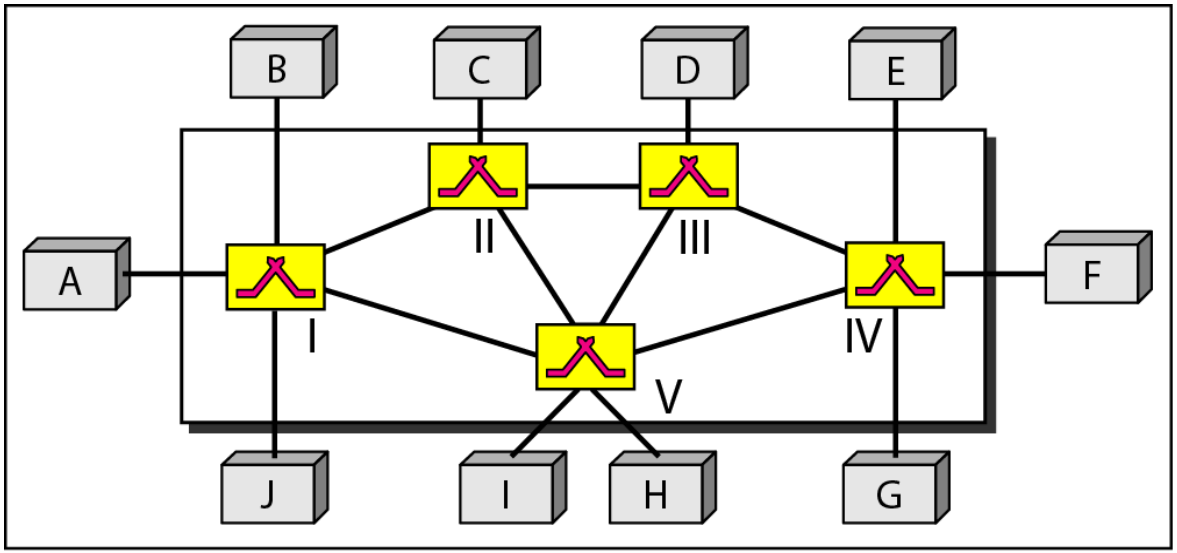
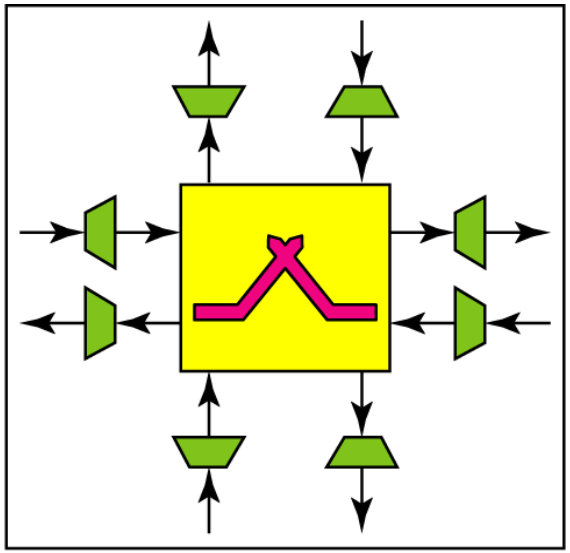


Figure 17.24 *A mesh SONET network*



a. SONET mesh network



b. Cross-connect switch

THE END

https://www.cisco.com/c/en/us/tech/optical/synchronous-optical-network-sonet/index.html